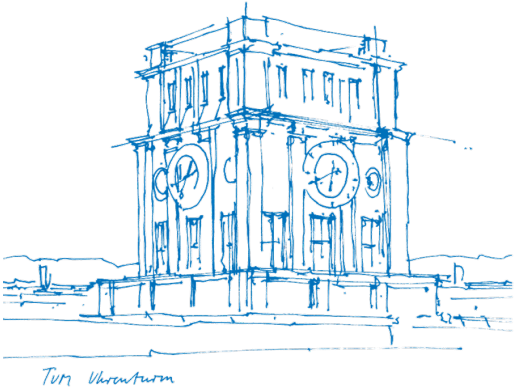
Coding-theoretic Challenges for DNA-based Storage/Communication Systems



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2023 6G-life Workshop on Molecular Communications

Introduction

DNA for storage systems

- ***Dense***: 1 Exabyte/mm^3 (5.5 Petabit//mm^3 in proof of concept (Church et al., 2012))
- ***Durable***: some mammoth DNA is still recoverable today!
- ***Timely***: never out of style (due to medical / genealogical applications)

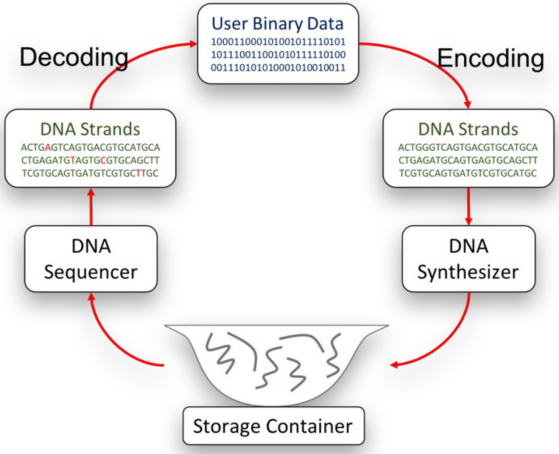
However...

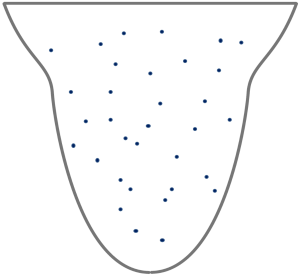
- Costly (although ever becoming more affordable):
  Synthesis for $4.5K per MB with Twist/Agilent (*Sequencing* for $0.26 per MB with Illumina Hiseq)

(figures courtesy of Technion DNAstorage lab)

- Also time consuming
- High error rate / abundant synchronization errors (deletions / insertion)

Life cycle of DNA-based storage system





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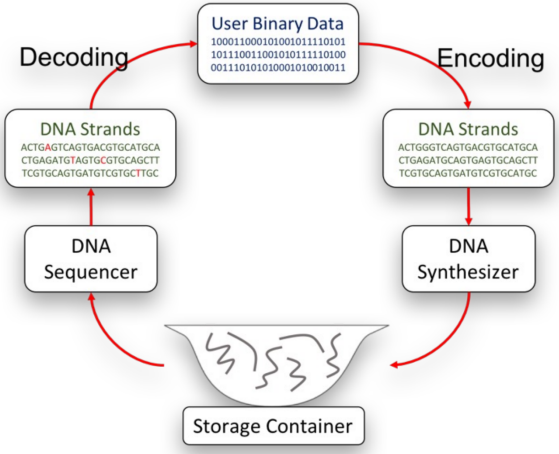

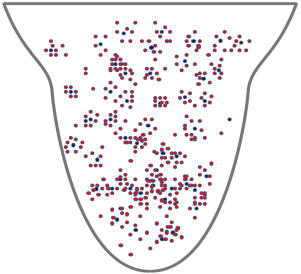

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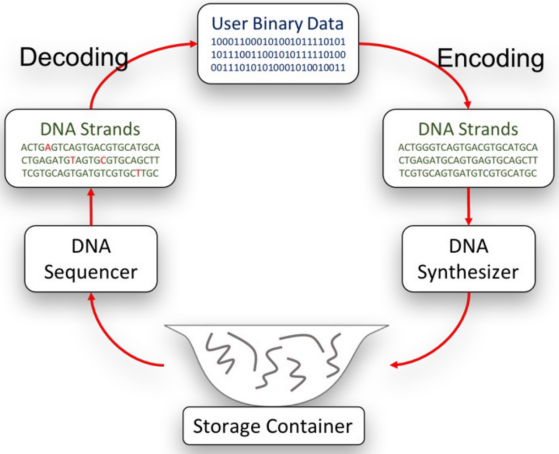

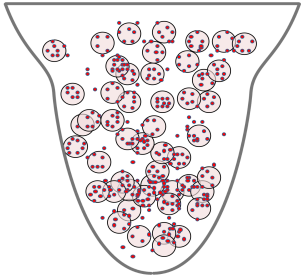

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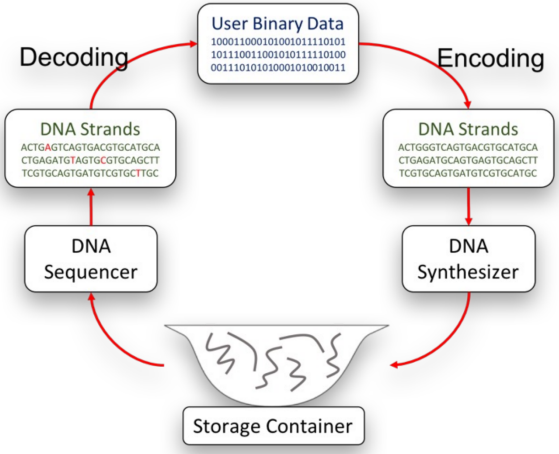

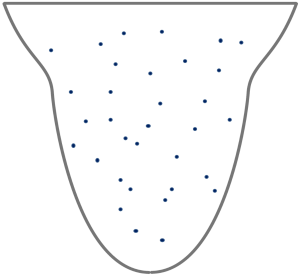

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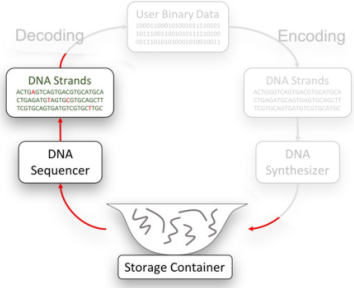
Lorenz

Yoni

DNA Sequencing

- Need for cheap, fast, accurate sequencers.

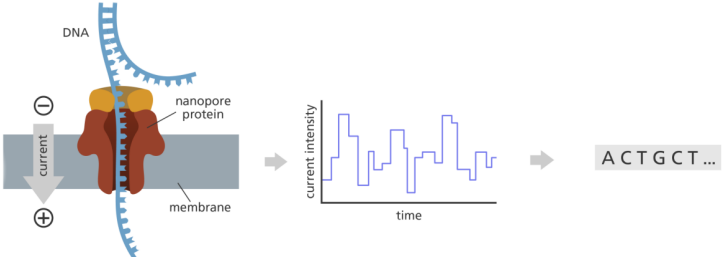
	Illumina	Nanopore
Read length (bases)	2-300	10-100K
Error rate	1%	10-15%
Cost	$2500 for 200M strands	$1000 for ≈50M strands



Source: *Bar-Lev, Sabary and Yaakobi, "Coding and Algoritms for DNA Storage Systems, ISIT'23 tutorial*

- Goals → Understanding capacity of the nanopore sequencer.
  → Design error-correcting codes tailored for it.

Nanopore Sequencing



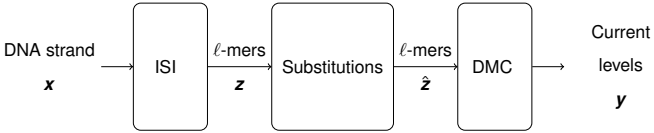
Source: *"What is Oxford Nanopore Technology (ONT) sequencing?," https://www.yourgenome.org/facts/what-is-oxford-nanopore-technology-ont-sequencing/*

- Sources of noise [MDK18]:
  - Nanopore holds $\ell > 1$ nucleotides at a time $\rightarrow$ Intersymbol interference (ISI)!
  - Strand moves irregularly $\rightarrow$ backtracking & skipping.
  - Noisy measurements

Prior Work

- [MDK18] → Introduced a mathematical model.
  → Established bounds on capacity.
- [HCW21] → Algorithm to compute capacity of an abstracted, deterministic channel model.
  → Coding schemes.
- [MVS22] → Finite-state Markov channel model incorporating major noise sources.
- [BYWY23] → Coding scheme for simplified model.

Simplified Model



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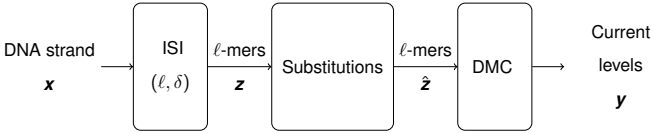

We model

- ISI as (ℓ, δ)-sliding window
- Measurement noise as substitutions

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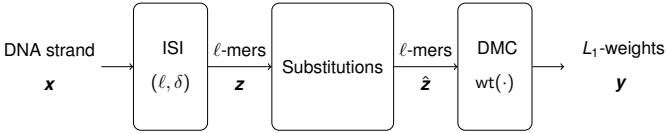

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Assumptions:

- Input $\boldsymbol{x} \in \{0, 1\}^n$
- Output: L_1-weights of ℓ-mers

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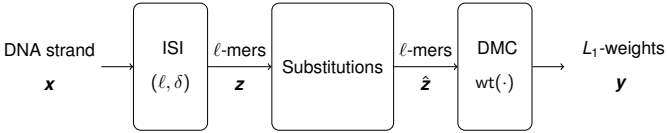

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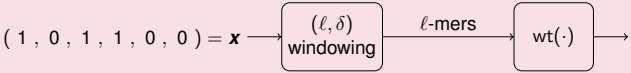

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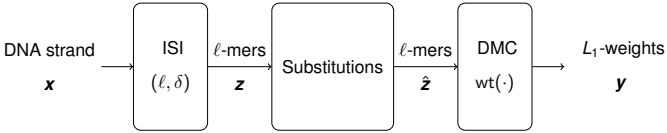

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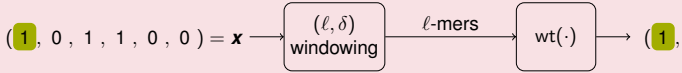

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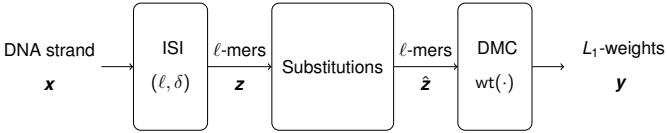

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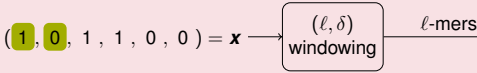

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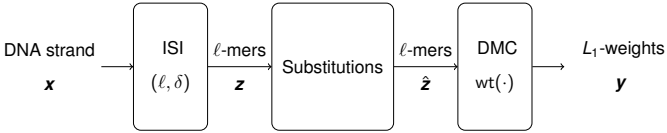

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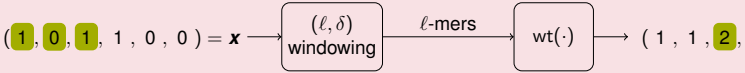

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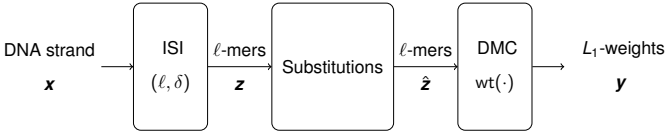

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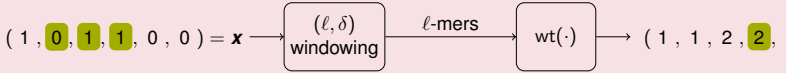

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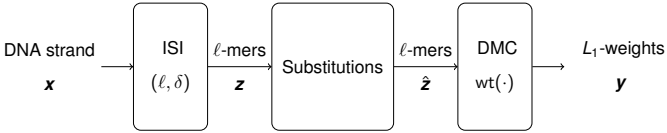

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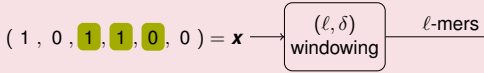

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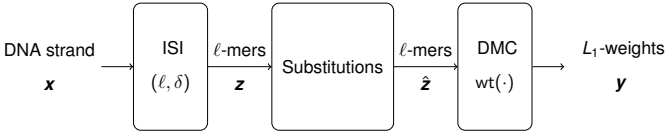

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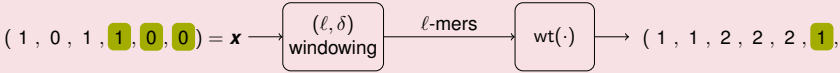

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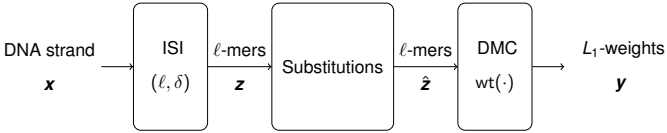

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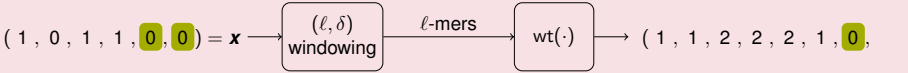

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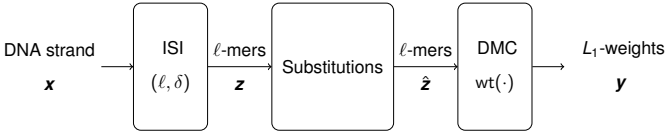

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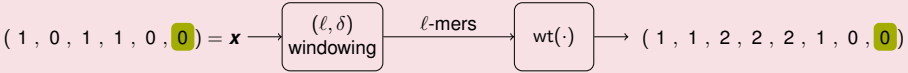

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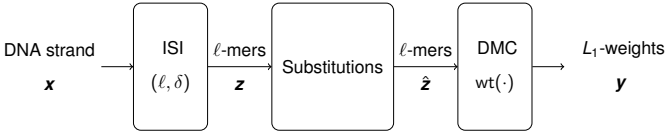

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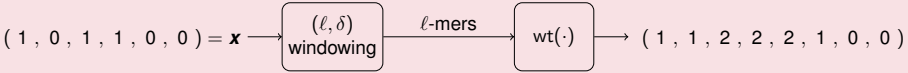

Read Vector

Definition

The (ℓ, δ)-*read vector* of any $\boldsymbol{x} \in \Sigma_2^n$ is of length $(n+\ell)/\delta - 1$ and denoted by

$$\mathcal{R}_{\ell,\delta}(\boldsymbol{x}) = (\mathrm{wt}(\boldsymbol{x}_{\delta-\ell+1}^{\delta}), \mathrm{wt}(\boldsymbol{x}_{2\delta-\ell+1}^{2\delta}), \ldots, \mathrm{wt}(\boldsymbol{x}_{n-\delta+1}^{n+\ell-\delta})),$$

where for $i \notin [n]$, let $x_i = 0$ and $\mathcal{R}_{\ell,\delta}(\boldsymbol{x})_i = \mathrm{wt}(\boldsymbol{x}_{i\delta-\ell+1}^{i\delta})$.

- *Aim*: Correct errors occurring in (ℓ, δ)-read vectors.

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Error-correcting Code

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$\mathcal{C}$ is a t-substitution $(\ell, 1)$-read code if for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{C}$,

$$d_H\big(\mathcal{R}(\boldsymbol{x}), \mathcal{R}(\boldsymbol{y})\big) > 2t,$$

where $d_H(\cdot, \cdot)$ denotes Hamming distance.

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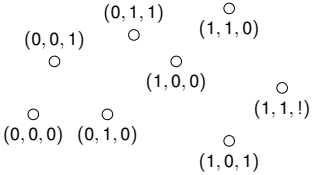
Results

- $\log\log n - o(1)$ min redundancy to correct 1 substitution in $(\ell \geqslant 3, \delta = 1)$-read vectors
- A 1-substitution correcting code that is optimal up to a constant.

Minimum Redundancy

- Consider graph with all vertices in $\{0,1\}^n$.



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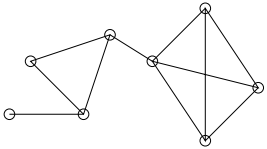

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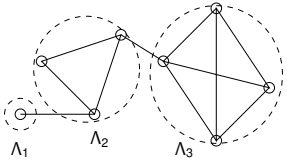
  - Clique $\Lambda \subset \{0,1\}^n \rightarrow$ all vertices pairwise-adjacent.



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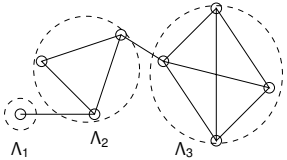
  - $\boldsymbol{x}, \boldsymbol{y} \in \{0,1\}^n$ adjacent iff $d_H(\mathcal{R}(\boldsymbol{x}), \mathcal{R}(\boldsymbol{y})) \leqslant 2$.
  - Clique $\Lambda \subset \{0,1\}^n \rightarrow$ all vertices pairwise-adjacent.
  - Clique cover $\mathcal{Q} \rightarrow$ collection of cliques that covers $\{0,1\}^n$.



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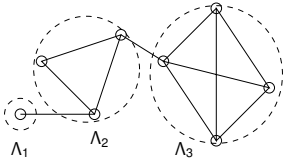

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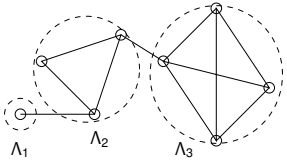

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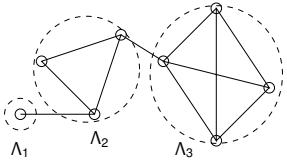

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Theorem

The redundancy of a 1-substitution $(\ell, 1)$-read code is bounded from below by

$$\log_2 \log_2(n) - o(1).$$

In Summary

- This or that

Thank you!